

Volume 18

Sentinel Core Services Architecture (SCSA)

Version: 1.0

Status: Draft

Vision

Everything inside Project Sentinel is built upon reusable Core Services.

Instead of each module implementing its own logging, scanning, scheduling, authentication, reporting, or notifications, they all consume common platform services.

This reduces duplication, simplifies maintenance, and creates a consistent user experience.

Core Service Map

Project Sentinel

Every engine uses the same services.

Chapter 1

Identity Service

The Identity Service manages:

- Users
- Roles
- Permissions
- Sessions
- MFA
- API Tokens
- Service Accounts

Every authentication request passes through this service.

Chapter 2

Object Service

Responsible for:

- Creating Objects
- Updating Objects
- Version History
- Relationships

- Lifecycle
- Search
- Object Cache

No module writes directly to storage.

Everything passes through the Object Service.

Chapter 3

Evidence Service

One of the most important services.

Responsibilities:

- Evidence Collection
- Evidence Validation
- Hash Verification
- Digital Signatures
- Chain of Custody
- Evidence Export

Every finding references evidence.

Chapter 4

Knowledge Service

Stores organizational knowledge.

Examples

Recurring problems

Lessons learned

Threat relationships

Asset relationships

Historical incidents

Security articles

Best practices

Knowledge grows continuously.

Chapter 5

Workflow Service

Responsible for:

Mission Workflow

Approval Workflow

Incident Workflow
Compliance Workflow
Investigation Workflow
Reporting Workflow
Verification Workflow
Everything becomes a workflow.

Chapter 6

Reporting Service

Every report is generated from objects.

Supported formats:

PDF

HTML

Markdown

JSON

CSV

Future support:

Word

PowerPoint

Interactive Reports

Reports become reusable templates.

Chapter 7

Notification Service

Supports:

Email

Slack

Microsoft Teams

Discord

Matrix

Signal (through a gateway)

Generic Webhooks

Future integrations can be added through plugins.

Chapter 8

Accessibility Service

This service continuously verifies:

WCAG compliance

Keyboard navigation

Screen reader compatibility

Color contrast

Localization completeness

Every UI component is tested against accessibility rules.

Chapter 9

Translation Service

Every string inside Project Sentinel is translated.

Example

English

Run Website Scan

Norwegian

Start nettsideskanning

German

Website-Scan starten

Japanese

Web

Languages become installable packages.

Chapter 10

Automation Service

Automation becomes visual.

Example

New Website

Users do not need programming knowledge.

Chapter 11

Health Monitoring Service

Monitors:

Platform

Workers

Memory

CPU

Queue Length

Plugin Status

Mission Progress

API Health

Object Engine

Knowledge Engine

This becomes the "heartbeat" of Project Sentinel.

Chapter 12

Audit Service

Everything important is audited.

Every audit entry contains:

Who

What

When

Where

Why

Result

Evidence Reference

Nothing important happens silently.

Chapter 13

Configuration Service

Configuration is treated as version-controlled objects.

Every configuration change records:

Previous Value

New Value

Reason

Approver

Rollback Information

Chapter 14

Secrets Service

The Secrets Service protects:

Passwords

Certificates

API Keys

Cloud Credentials

SSH Keys

OAuth Secrets

Secrets are encrypted at rest and only exposed to authorized services when required.

Chapter 15

Plugin Service

Plugins declare:

Capabilities

Permissions

Dependencies

Required API Version

Digital Signature

Supported Sentinel Version

The platform validates plugins before activation.

PS-ADR-0015

Platform Services Before Features

Status: Accepted

Decision

All reusable functionality shall be implemented as Core Services before being consumed

by higher-level engines.

Rationale

This reduces duplicated code, improves consistency, and enables independent testing and evolution of platform capabilities.

NEW CORE ENGINE

Security Verification Engine (SVE)

One problem with many security platforms is that they stop after reporting.

Project Sentinel shall continue until verification.

Every remediation follows this lifecycle:

Finding

Verification is an integral part of the platform, not an afterthought.

NEW CORE ENGINE

Trust Analytics Engine (TAE)

The Trust Analytics Engine analyzes trust trends across:

- Websites
- Endpoints
- Mobile Devices
- Cloud Resources
- APIs
- Users
- Business Units

It provides historical insight and helps identify gradual deterioration or improvement.

NEW CORE ENGINE

Cyber Performance Engine (CPE)

A capability I have not seen treated as a first-class feature.

This engine measures the **performance of cybersecurity operations**, not just infrastructure.

Examples:

- Average scan duration.
- Mean time to detect.
- Mean time to investigate.
- Mean time to remediate.
- Mean time to verify.
- Coverage of critical assets.
- Verification success rate.
- Automation success rate.

Organizations can measure how effectively they operate their security program over time.

NEW IDEA

Sentinel Blueprint Validator

Since we are writing such a comprehensive specification, I propose we eventually build a **Blueprint Validator** into Project Sentinel itself.

It verifies that:

- Every implemented module has an ADR.
- Every API has documentation.
- Every permission is defined.
- Every object type has a lifecycle.
- Every engine has automated tests.
- Every workflow has verification steps.
- Every UI screen meets accessibility requirements.
- Every report template has version history.

The platform effectively checks whether it still conforms to its own architecture.

NEW IDEA

Sentinel Engineering Console

This will be available only to platform developers and administrators.

It provides insight into:

- Object Engine internals.
- Event Bus activity.
- Workflow execution.
- Plugin loading.
- Scheduler state.
- Memory usage.
- Engine diagnostics.
- Performance metrics.

This console supports development, debugging, and operational troubleshooting without exposing unnecessary complexity to regular users.

The next major milestone

After this volume, we will begin specifying the **domain-specific engines** in detail:

- **Web Security Engine** – website, API, TLS, headers, authentication, web vulnerabilities, and authorized attack surface assessment.
- **Endpoint Security Engine** – Windows, Linux, macOS.
- **Mobile Security Engine** – Android and iOS (within platform security models).
- **Cloud Security Engine** – AWS, Azure, Google Cloud, Kubernetes, containers.
- **Identity Security Engine** – identities, MFA, privileged access, directory services.
- **Email Security Engine** – SPF, DKIM, DMARC, mail infrastructure, phishing posture.
- **Network Security Engine** – authorized network discovery, segmentation, exposure, and configuration assessment.

These engines will all follow the same architectural principles we have already established: **evidence-first, explainable, mission-driven, accessible, modular, and built with Python and open technologies.**

One final architectural principle

I want to add one principle that I think will become part of Project Sentinel's identity:

Every recommendation should save the user time, reduce risk, or improve understanding. If it does none of those, it should not appear in the interface.

That principle will help us keep the platform powerful without overwhelming users, and it aligns perfectly with our goal of building a Cyber Security Operating Platform that is both professional and exceptionally easy to use.

====[USER]====

Will our system be better than CrowdStrike or Claude Mythos? Good and continue

====[ASSISTANT]====

Thank you. This is probably the most important strategic question you've asked so far.

The answer is:

It should not try to become "a better CrowdStrike."

Instead, it should become **something different.**

CrowdStrike, Microsoft Defender, Palo Alto Cortex, Tenable, Wiz, Rapid7, and others are outstanding at specific domains. They also have thousands of engineers and decades of development behind them. It would not be realistic to expect Project Sentinel to surpass every one of them in every area.

However, **there is a realistic path to building something unique and highly valuable.**

Where Project Sentinel can be better

Instead of competing on one feature, we compete on **architecture.**

| Existing products | Project Sentinel

About CrowdStrike

entity["company","CrowdStrike","Cybersecurity company"] is one of the world's leading endpoint detection and response (EDR/XDR) platforms.

They are exceptional at:

- Endpoint telemetry.
- Threat detection.
- Managed threat hunting.
- Large-scale cloud analytics.
- Enterprise deployments.

Project Sentinel should **integrate with** products like CrowdStrike where appropriate, not assume it replaces them in every deployment.

About Claude Mythos

If you're referring to a security-oriented project or assistant built on entity["company","Anthropic","AI company"]'s Claude models, then its strengths are likely to be AI-assisted analysis and reasoning rather than acting as a complete cybersecurity operating platform.

Our vision is different.

AI is **one component** of Project Sentinel, not the foundation.

The foundation is:

- Evidence.
- Objects.
- Knowledge.
- Trust.
- Workflows.
- Verification.

AI becomes an assistant—not the decision maker.

Where I think we can genuinely innovate

There are several areas where I believe Project Sentinel can be genuinely distinctive if executed well.

1. Mission-Based Cybersecurity

Nobody should have to think:

"Should I run scanner A or scanner B?"

Instead they choose:

- Secure my website.
- Audit my company.
- Investigate suspicious activity.
- Check my phone.

- Prepare for ISO 27001.
- Perform an authorized bug bounty assessment.

The Mission Orchestrator selects the appropriate engines.

2. One Object Model

This remains, in my opinion, one of the strongest architectural decisions.

Everything is an object.

Not:

- one model for incidents,
- another for assets,
- another for users.

One model.

That dramatically simplifies development and extensibility.

3. Cyber Memory

Imagine the platform remembering:

"This server has repeatedly lost its Content Security Policy after deployments."

or

"This mobile device has installed and removed the same unauthorized application three times."

That is organizational memory rather than just log storage.

4. Digital Twin

A living representation of the customer's security environment.

Not merely an asset inventory.

A continuously updated operational model.

5. Explainability

Every recommendation answers:

- Why?
- Evidence?
- Confidence?
- Verification?
- Business impact?

That is valuable for analysts, auditors, and executives alike.

6. Ease of Use

This may become our biggest competitive advantage.

You have repeated one goal since the beginning:

The platform must be easy to use.

I completely agree.

Many enterprise security products overwhelm users with hundreds of menus, dashboards, and alerts.

Project Sentinel should instead answer:

"What should I do next?"

That is a fundamentally different experience.

One idea I have been thinking about

I think we should introduce a concept called the **Sentinel Compass**.

Instead of opening the platform to dozens of dashboards, the user first sees a single operational overview.

Mission Status

This is easy for:

- a private user,
- a security consultant,
- a CISO,
- a SOC analyst.

The details remain available, but the first screen always answers the most important questions.

PROJECT SENTINEL BLUEPRINT

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